

Manual Muscle Testing

Cervical Spine

Cervical Flexion

- Muscles involved
 - Sternocleidomastoid
 - Longus colli
 - Scalene anterior
 - Scalenus medius
 - Sternohyoid
 - Omohyoid
- ROM
 - 0 degree to 35-45 degrees



Cervical Flexion

- Grade 5 and Grade 4
 - Patient's position – Supine with arms at side. Head supported on table.
 - Therapist's position – Standing next to patient's head. Hand for the resistance will be placed on forehead and other hand can be on the chest.
 - Procedure – Patient flexes neck straight up without tucking the chin and then therapist gives resistance through the forehead.
 - Conclusion
 - Grade 5 – Patient completes available ROM against max resistance.
 - Grade 4 – Patient completes the available ROM against moderate resistance.

Cervical Flexion

- Grade 5 and Grade 4

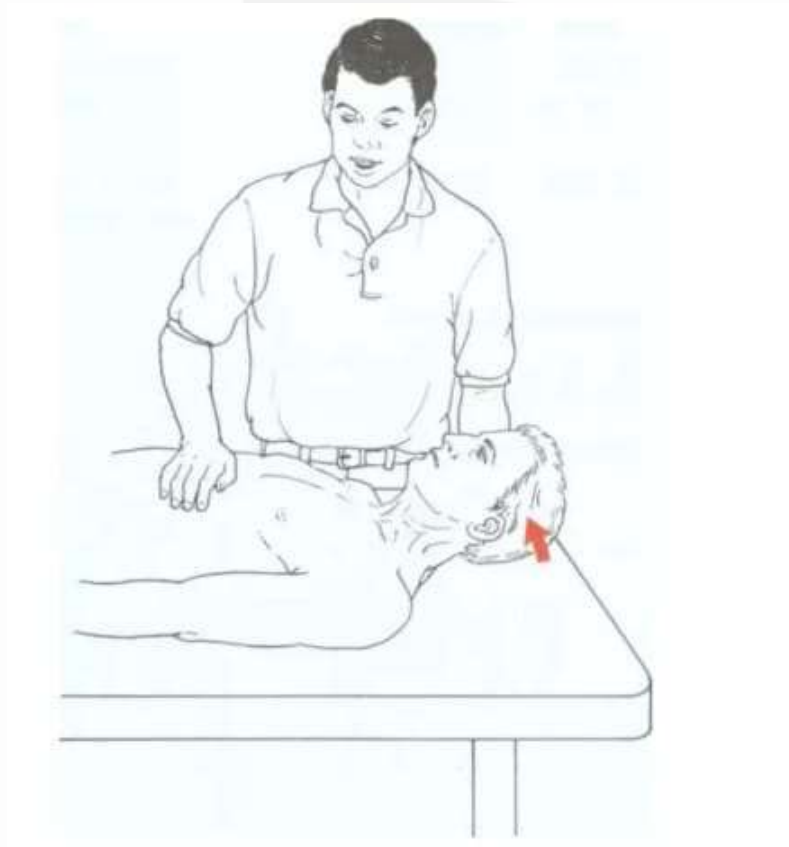


Cervical Flexion

- Grade 3
 - Patient's position – Supine with arms at side. Head supported on table.
 - Therapist's position – Standing next to patient's head.
 - Procedure – Patient flexes the neck into the complete ROM without any external resistance and against the gravity.
 - Conclusion
 - Grade 3 – Patient completes available ROM against max resistance.

Cervical Flexion

- Grade 3



Cervical Flexion

- Grade 2, Grade 1 and Grade 0
 - Patient's position – Supine with arms at side. Head supported on table.
 - Therapist's position – Standing at the head of the table facing the patient. Finger of both hands over the sternocleidomastoid to palpate them during the test.
 - Procedure – Patient is instructed to roll his/her head to the left and to the right.
 - Conclusion
 - Grade 2 – Patient completes partial ROM. (Right sternocleidomastoid produces roll to the left side and vice versa)
 - Grade 1 – No motion occurs but contractile activity in one or both muscle is detected.
 - Grade 0 – No motion and No contractile activity

Cervical Extension

- Muscles
 - Longissimus cervicis
 - Semispinalis cervicis
 - Iliocostalis cervicis
 - Upper Trapezius



Cervical Extension

- Grade 5 and Grade 4
 - Patient's position – Prone with head off end of the table. Arms at side.
 - Therapist's position – Standing next to patient's head. One hand is placed at the parietooccipital area to give resistance (downward & forward). Other hand below chin.
 - Procedure – Patient is advised to lift up the head and look at the ceiling. Therapist will resist the movement.
 - Conclusion
 - Grade 5 – Patient completes ROM under maximum resistance
 - Grade 4 – Patient completes ROM under moderate resistance

Cervical Extension

- Grade 5 and Grade 4



Cervical Extension

- Grade 3
 - Patient's position – Prone with head off end of the table. Arms at side.
 - Therapist's position – Standing next to patient's head.
 - Procedure – Patient is advised to raise up the head.
 - Conclusion
 - Grade 3 – Patient completes ROM without resistance.

Cervical Extension

- Grade 3



Cervical Extension

- Grade 2, Grade 1 and Grade 0
 - Patient's position – Prone with head fully supported on the table. Arms at side.
 - Therapist's position – Standing next to patient's upper trunk. Both hands on cervical region and base of occiput for palpation.
 - Procedure – Patient is advised to raise up the head and look up.
 - Conclusion
 - Grade 2 – Patient moves through partial ROM.
 - Grade 1 – Contractile activity felt in the extensor muscles but no movement.
 - Grade 0 – No contractile activity

Cervical Rotation

- Grade 5, Grade 4 and Grade 3
 - **Patient's position** – Supine with cervical spine in neutral. Head supported on table with head turned to one side.
 - **Therapist's position** – Standing at head of table facing patient. Hand for resistance is placed over the side of head above ear.
 - **Procedure** – Patient is advised to rotate his/her head back to the neutral position against resistance.
 - **Conclusion**
 - **Grade 5** – Patient rotates head through full ROM against maximal resistance.
 - **Grade 4** – Patient rotates head through full ROM against moderate resistance.
 - **Grade 3** – Patient rotates head through full ROM without any resistance.

Cervical Rotation

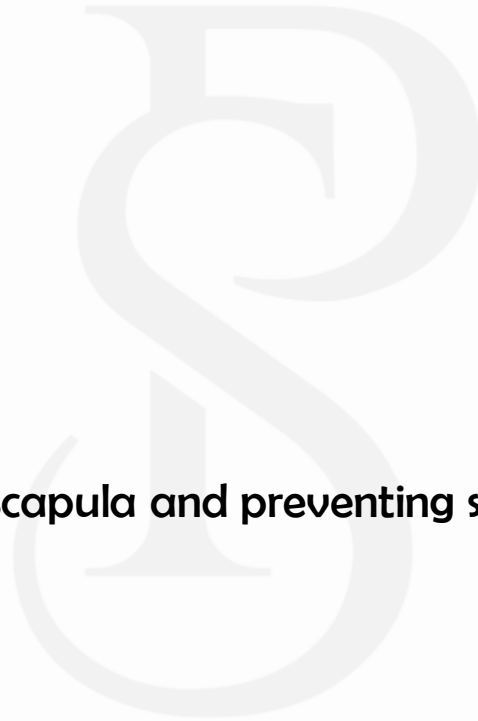
- Grade 2, Grade 1 and Grade 0
 - **Patient's position** – Sitting with trunk and head may be supported against a high-back chair. Head should be neutral.
 - **Therapist's position** – Standing in front of the patient.
 - **Procedure** – Patient is advised to rotate his/her head to both left and right from the neutral position.
 - **Conclusion**
 - **Grade 5** – Patient completes partial ROM.
 - **Grade 4** – Contractile activity is seen in the muscles.
 - **Grade 3** – No contraction/movement.

Manual Muscle Testing

Shoulder Joint

Shoulder Flexion

- **Muscles involved**
 - Anterior Deltoid
 - Coracobrachialis
- **Accessory Muscles**
 - Pectoralis Major (Upper)
 - Deltoid (Middle)
 - Serratus anterior (Upwardly rotating scapula and preventing scapular adduction)



Shoulder Flexion

- Grade 5 and Grade 4
 - Patient's position – Short sitting with arms at sides, elbow slightly flexed, forearm pronated.
 - Therapist's position – Stand at test side. Hand will give resistance is placed over the distal humerus just above the elbow. The other hand will stabilize the shoulder.
 - Procedure – Ask the patient to flex the shoulder to 90 degree without rotation or horizontal movement.
 - Conclusion
 - Grade 5 – Hold the end position against max resistance.
 - Grade 4 – Hold the end position against moderate resistance.

Shoulder Flexion

- Grade 5 and Grade 4

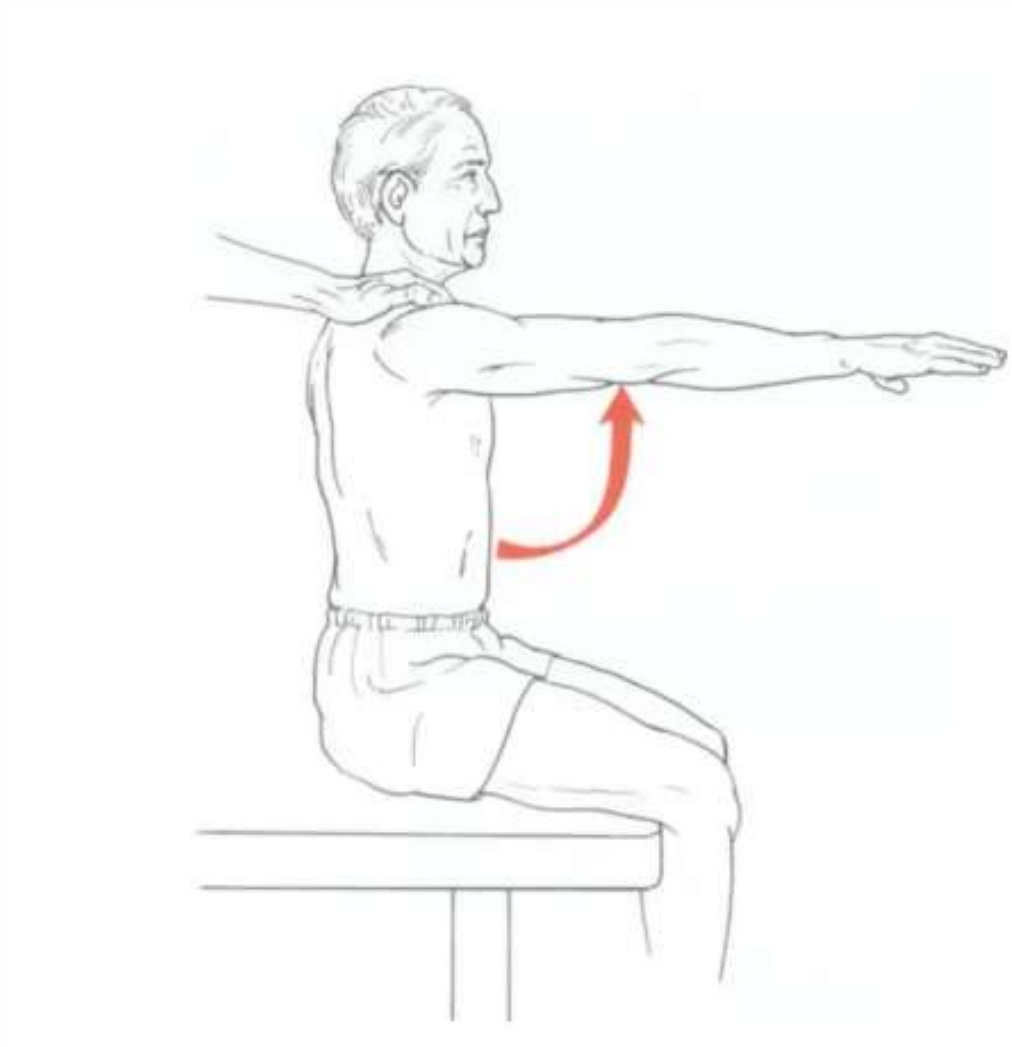


Shoulder Flexion

- Grade 3
 - Patient's position – Short sitting with arms at sides, elbow slightly flexed, forearm pronated.
 - Therapist's position – Stand at test side
 - Procedure – Ask the patient to flex the shoulder to 90 degree.
 - Conclusion
 - Grade 3 – Patient completes available ROM against the gravity but without any resistance.

Shoulder Flexion

- Grade 3



Shoulder Flexion

- Grade 2, Grade 1 and Grade 0
 - Patient's position – Short sitting with arms at sides, elbow slightly flexed
 - Therapist's position – Stand at test side. Fingers is used for palpation are placed on the anterior Deltoid over the shoulder joint.
 - Procedure – Ask the patient to flex the shoulder to 90 degree.
 - Conclusion
 - Grade 2 – Patient completes partial ROM
 - Grade 1 – No motion occurs but contractile activity in the deltoid is detected.
 - Grade 0 – No motion and No contractile activity

Shoulder Flexion

- Grade 2, Grade 1 and Grade 0



Shoulder Extension

- Muscles involved
 - Latissimus Dorsi
 - Deltoid (Posterior)
 - Teres Major
- Accessory Muscles
 - Triceps brachii (Long head)

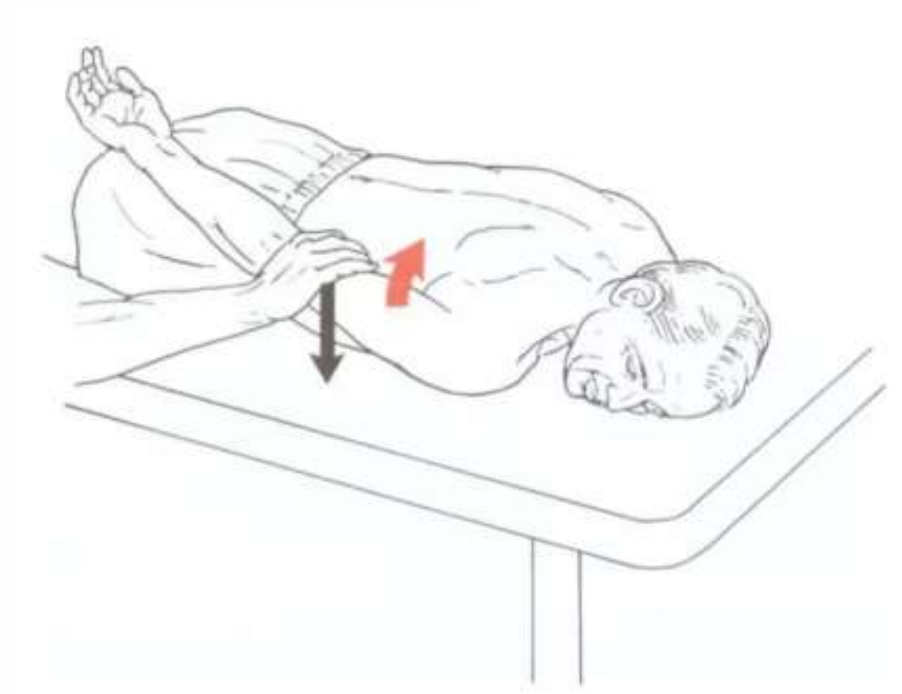
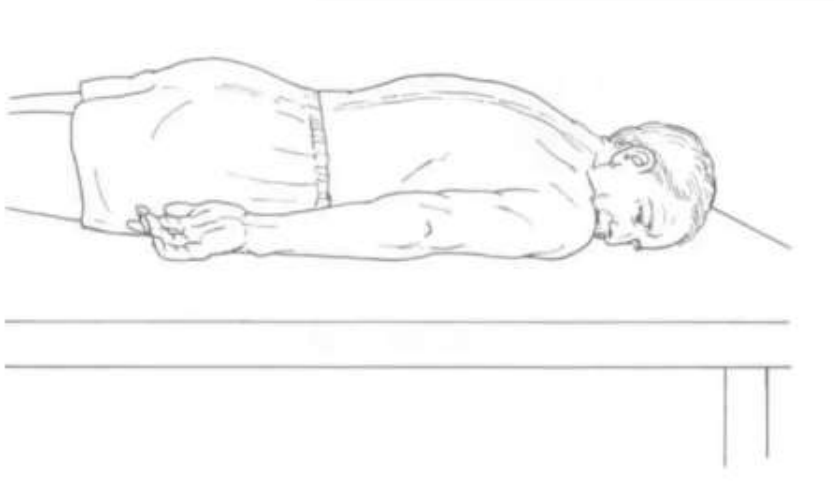


Shoulder Extension

- Grade 5 and Grade 4 (TEST 1)
 - Patient's position – Prone with arms at sides and shoulder internally rotated (palm up)
 - Therapist's position – Stand at test side. Hand used for resistance is placed over the posterior arm just above the elbow joint.
 - Procedure – Ask the patient to extend the shoulder keeping the elbow straight.
 - Conclusion
 - Grade 5 – Completes available ROM against max resistance.
 - Grade 4 – Completes available ROM against moderate resistance.

Shoulder Extension

- Grade 5 and Grade 4

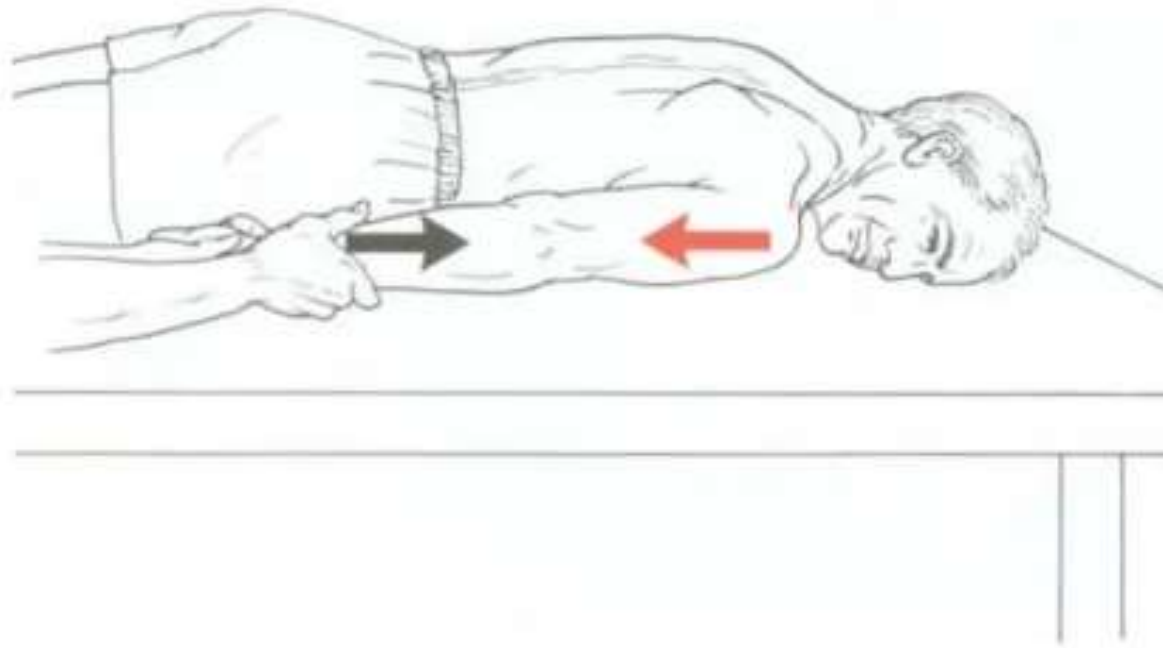


Shoulder Extension

- Grade 5 and Grade 4 (TEST 2 – To isolate Latissimus dorsi)
 - Patient's position – Prone with head turned to test side. Arms at sides and shoulder internally rotated. Test shoulder is hiked to the level of the chin.
 - Therapist's position – Stand at test side. Grasp the forearm above the patient's wrist with both hands
 - Procedure – Patient depresses arm caudally and therapist will push the arm upward towards the patient's head.
 - Conclusion
 - Grade 5 – Completes available ROM against max resistance.
 - Grade 4 – Completes available ROM against moderate resistance.

Shoulder Extension

- Grade 5 and Grade 4

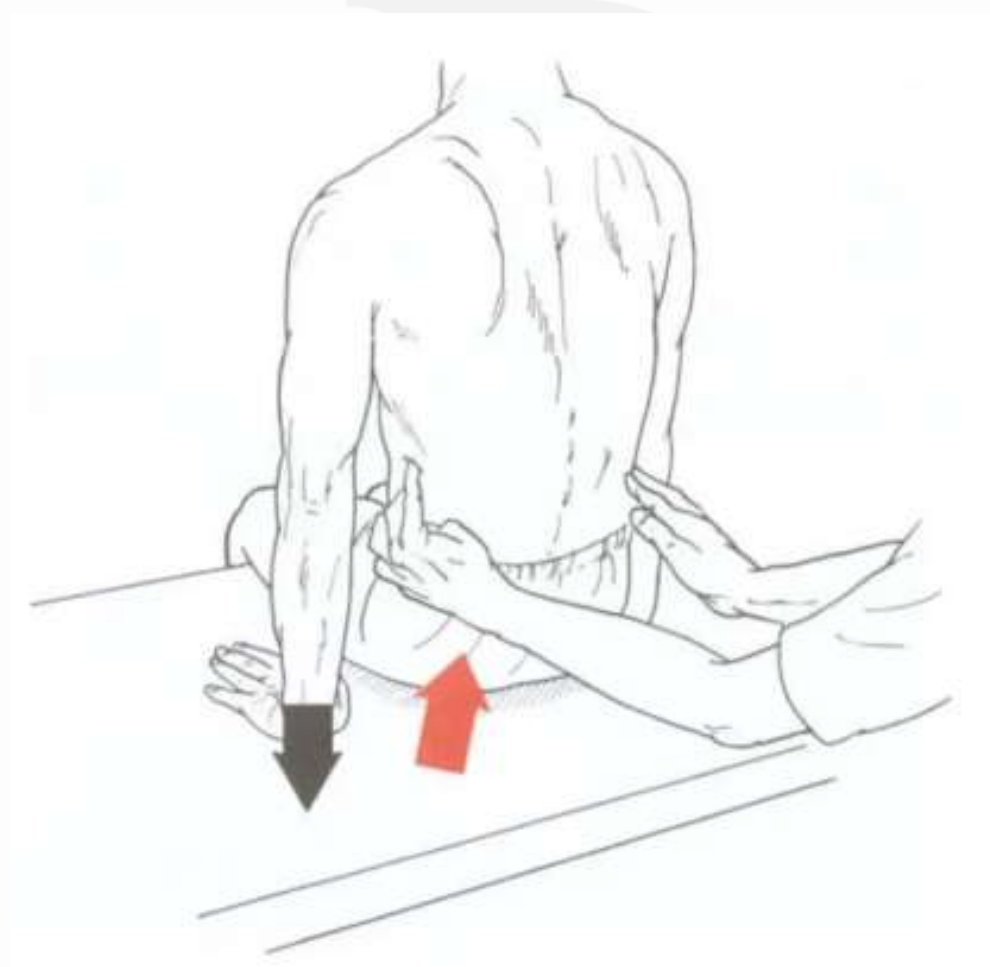


Shoulder Extension

- Grade 5 and Grade 4 (TEST 3 – To isolate Latissimus dorsi)
 - Patient's position – Short sitting with hands flat on the table adjacent to hips.
 - Therapist's position – Stand behind the patient. Fingers placed to palpate the latissimus dorsi fibres on the lateral aspect of the thoracic wall (B/L) just above the waist.
 - Procedure – Patient pushes the hand downward and lift buttocks from table.
 - Conclusion
 - Grade 5 – Patient is able to lift the buttocks completely from the table.
 - Grade 4 – Can't determined

Shoulder Extension

- Grade 5 and Grade 4

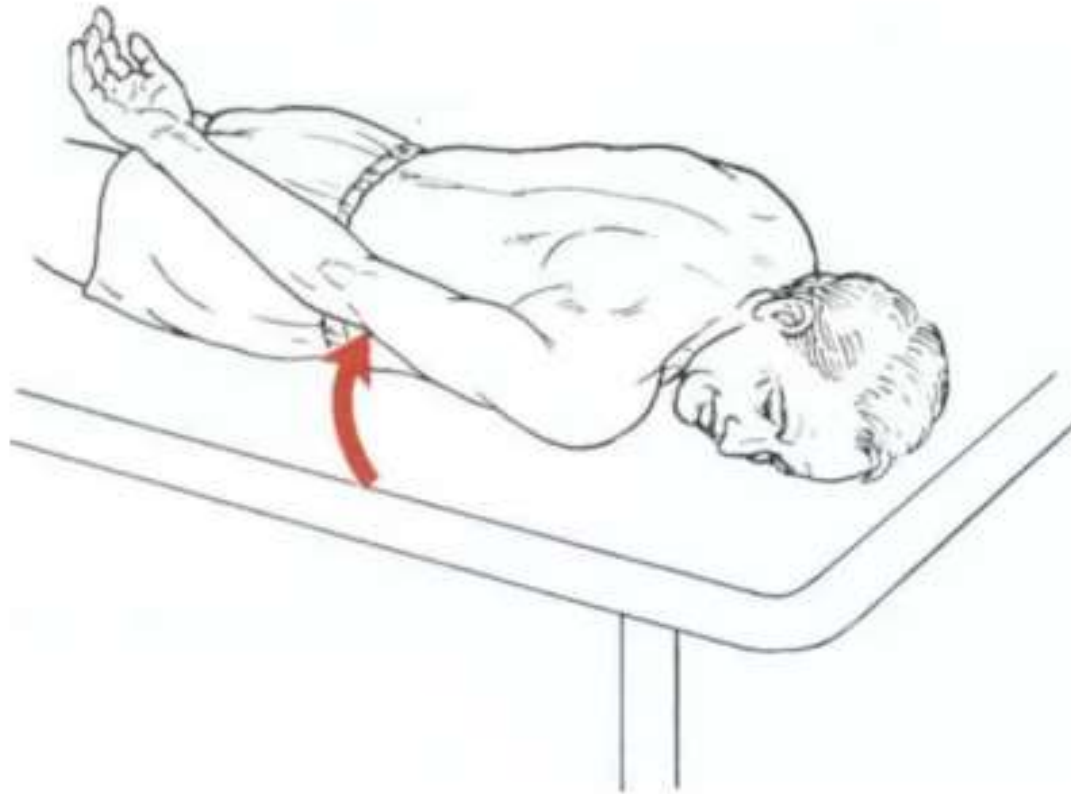


Shoulder Extension

- Grade 3 and Grade 2
 - Patient's position – Prone with head turned to one side. Arms at sides and test arm is internally rotated.
 - Therapist's position – Stand at the test side.
 - Procedure
 - Test 1 – Patient raises arm off the table
 - Test 2 (Latissimus) – Patient pushes arm towards the feet
 - Conclusion
 - Grade 3 – Complete available ROM without any manual resistance.
 - Grade 2 – Patient complete partial ROM

Shoulder Extension

- Grade 3 and Grade 2

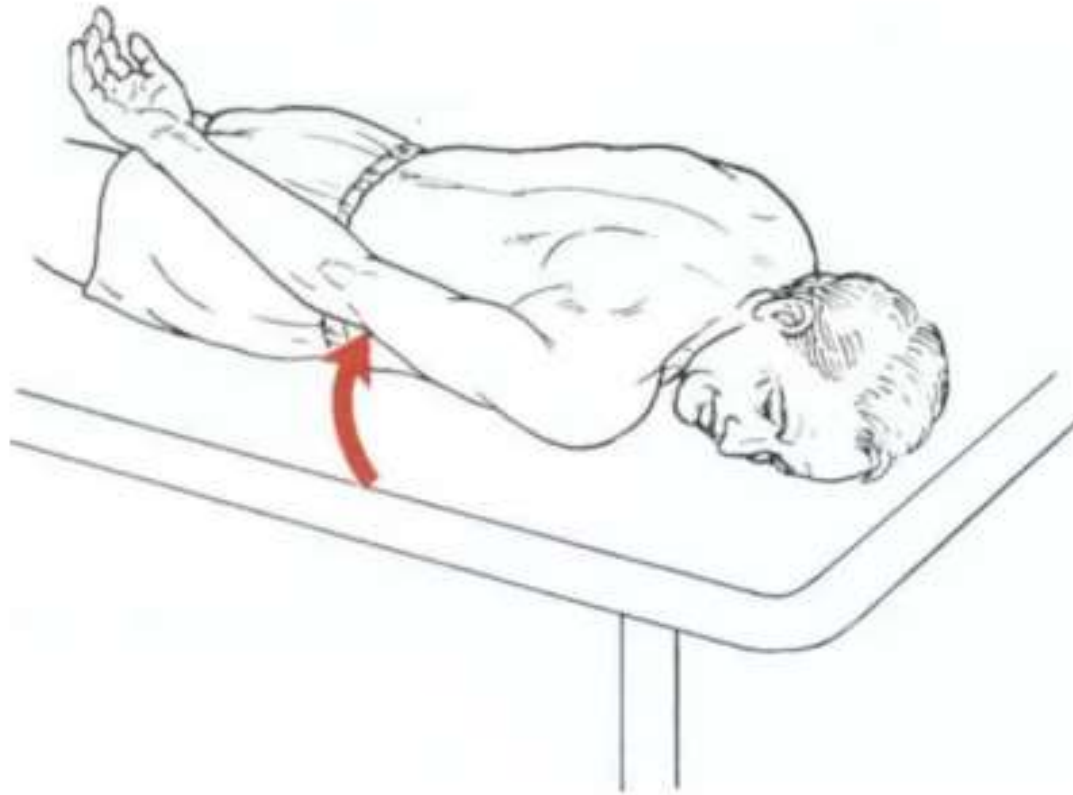


Shoulder Extension

- Grade 1 and Grade 0
 - Patient's position – Prone with arms at sides and shoulder internally rotated (Palm up).
 - Therapist's position – Stand at the test side. Fingers for palpation (Latissimus) are placed on the side of the thoracic wall below and lateral to the inferior angle of the scapula.
 - Procedure – Patient attempts to lift the arm from table.
 - Conclusion
 - Grade 2 – Palpable contractile activity in any of the participating muscles but no movement of the shoulder.
 - Grade 1 – No contractile response in the muscles

Shoulder Extension

- Grade 1 and Grade 0





Shoulder Abduction

- Muscles involved
 - Deltoid (Middle fibres)
 - Supraspinatus

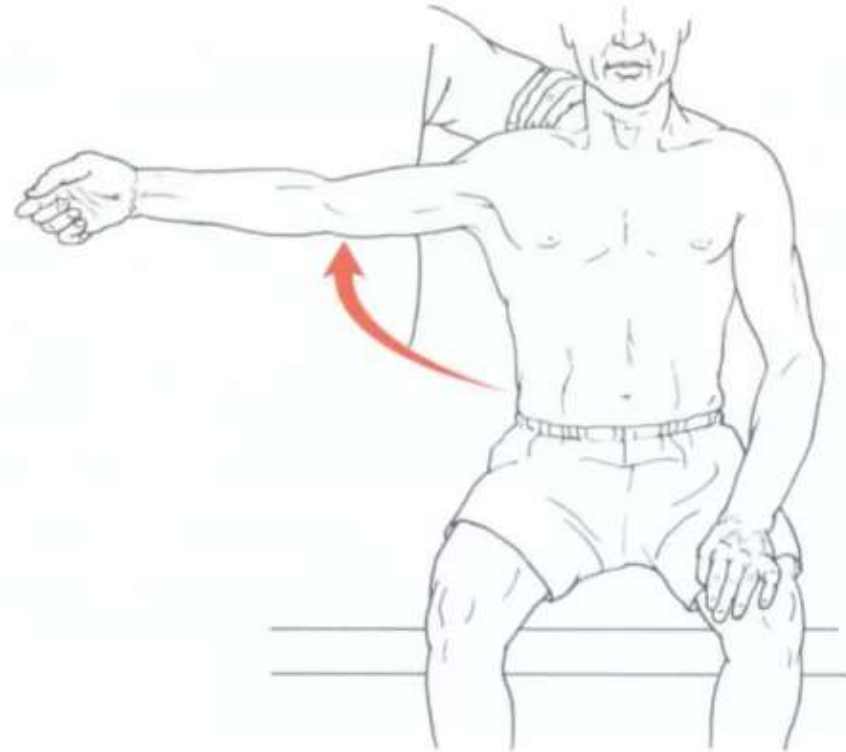


Shoulder Abduction

- Grade 5, Grade 4 & Grade 3
 - Patient's position – Short sitting with arm at side and elbow slightly flexed.
 - Therapist's position Stand behind patient. Hand giving resistance is placed over the arm just above elbow.
 - Procedure – Ask the patient to abduct the arm at 90 degrees.
 - Conclusion
 - Grade 5 – Holds end test position against max resistance.
 - Grade 4 – Holds end test position against moderate resistance.
 - Grade 3 – Completes range of motion to 90 degrees with no manual resistance.

Shoulder Abduction

- Grade 5. Grade 4 & Grade 3



Shoulder Abduction

- Grade 2

- **Patient's position** – Short sitting with arm at side and elbow slightly flexed.
- **Therapist's position** Stand behind patient. Palpate the deltoid lateral to the acromial process on the superior of the shoulder. The supraspinatus can be palpated by placing the fingers deep under the Trapezius in the supraspinous fossa.
- **Procedure** – Ask the patient to abduct the arm.
- **Conclusion**
 - **Grade 2** – Completes partial ROM

Shoulder Abduction

- Grade 2

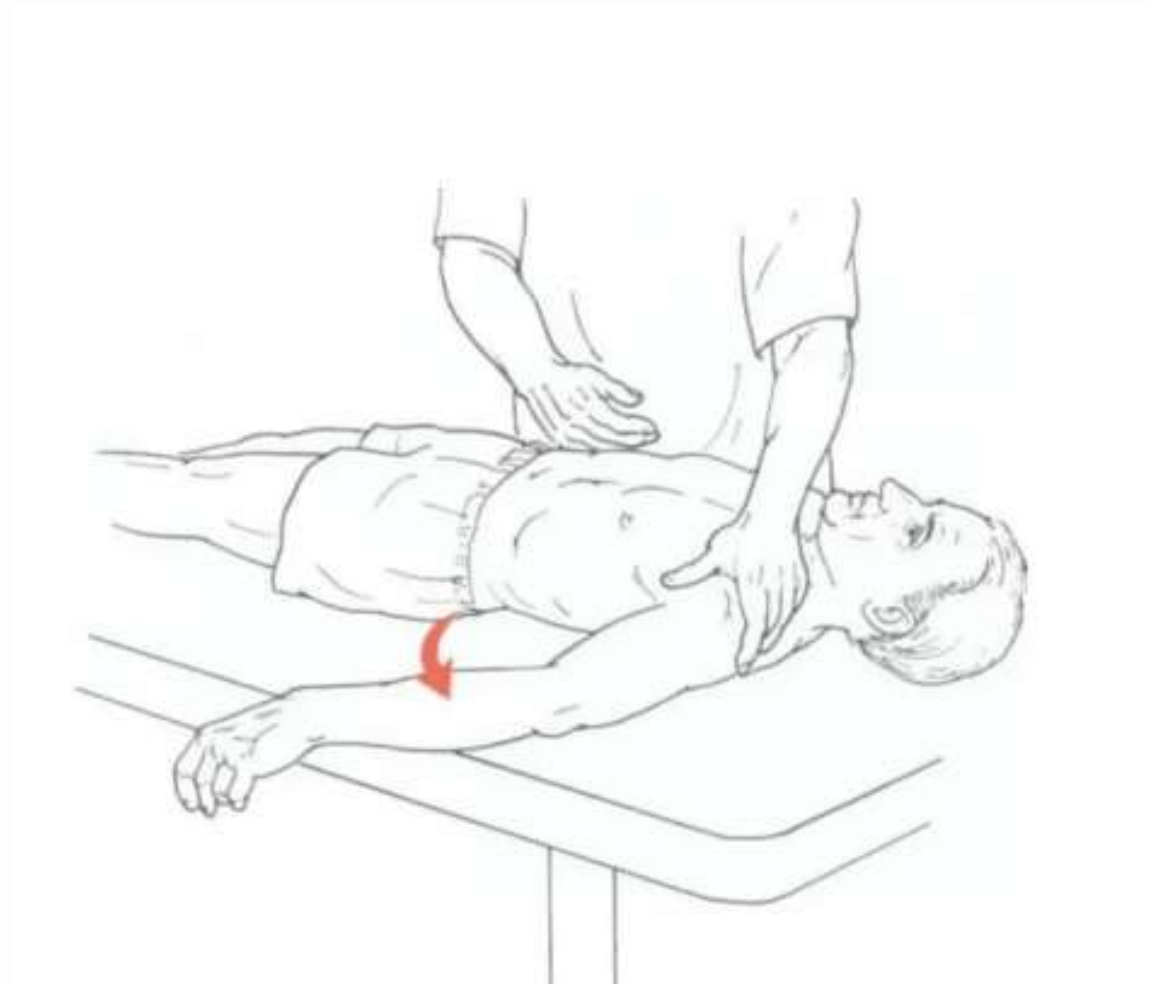


Shoulder Abduction

- Grade 2
 - Patient's position – Supine with arm at side supported on table
 - Therapist's position – Stand at side of patient. Palpate the deltoid lateral to the acromial process on the superior of the shoulder. The supraspinatus can be palpated by placing the fingers deep under the Trapezius in the supraspinous fossa.
 - Procedure – Ask the patient to abduct the arm.
 - Conclusion
 - Grade 2 – Completes full ROM

Shoulder Abduction

- Grade 2



Shoulder Abduction

- Grade 1 & Grade 0
 - Patient's position – Short sitting
 - Therapist's position – Stand behind and to the side of patient. Therapist cradles test arm with the shoulder in about 90 degree of abduction providing limb support at the elbow.
 - Procedure – Patient tries to maintain the arm in abduction
 - Conclusion
 - Grade 2 – Palpable or visible contraction of Deltoid with no movement.
 - Grade 1 – No contraction activity

Shoulder Abduction

- Grade 1 & Grade 0



Shoulder Abduction

- Grade 1 & Grade 0
 - Patient's position – Supine with arm at side and elbow slightly flexed.
 - Therapist's position – Stand at side of table at a place where Deltoid can be reached. Palpate the deltoid on the lateral surface of the upper 1/3rd of the arm.
 - Procedure – Ask the patient to abduct the arm.
 - Conclusion
 - Grade 2 – Palpable or visible contraction of Deltoid with no movement.
 - Grade 1 – No contraction activity

Shoulder Abduction

- Grade 1 & Grade 0



Shoulder Horizontal Abduction

- Grade 5, Grade 4 & Grade 3
 - **Patient's position** – Shoulder abducted to 90 degrees and forearm off the edge of the table with elbow flexed.
 - **Therapist's position** – Stand at side. Hand giving resistance is placed over the posterior aspect of arm just above the elbow.
 - **Procedure** – Ask the patient to abduct the arm horizontally.
 - **Conclusion**
 - **Grade 5** – Complete ROM and holds the position against maximum resistance
 - **Grade 4** – Complete ROM and holds the position against moderate resistance
 - **Grade 3** – Complete ROM with no manual resistance

Shoulder Horizontal Abduction

- Grade 5, Grade 4 & Grade 3



Shoulder Horizontal Abduction

- Grade 2, Grade 1 & Grade 0
 - Patient's position – Short sitting over end or side of the table
 - Therapist's position – Stand at test side. Support forearm under distal surface and palpate over the posterior surface of the shoulder just superior to the axilla.
 - Procedure – Ask the patient to abduct the arm horizontally.
 - Conclusion
 - Grade 2 – Moves partial ROM
 - Grade 1 – Muscle contraction and no Motion
 - Grade 0 – No contraction

Shoulder Horizontal Abduction

- Grade 5, Grade 4 & Grade 3

